12th BOARD HACKER JUGAAD NOTES:-

Probability:-

→ Its mathematical calculation of chance of occurence of an event E and it can be denoted as P(E).

$$0 \leq P(E) \leq 1$$

→ Also it lies in interval

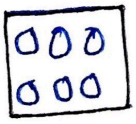
$[0,1]$

$0 \leqslant P(E) \leqslant 1$

Related terms:-

1. Impossible Event:-
It will have probability 0.

$P(E) = 0 \Rightarrow$ E can never occur



2. Sure Event:- It will have probability 1.

$P(E) = 1 \Rightarrow$ E will occur definitely.

3. Sample Point:- A possible outcome of an experiment is called sample point.

H → Sample point

T → Sample point

4. Sample Space:- It is a set containing all possible outcomes. It is a collection of all sample point.

If die is cast → sample space $\equiv \{1,2,3,4,5,6\}$

Sample space = {H, T}

5. Random Experiment:-
It is experiment for which all possible outcomes are known, but obtained outcome of a particular trial is not known in advance.

6. Event:- It is a subset of sample space.

Sample Space = {1, 2, 3, 4, 5, 6}

E_1 = No is even

E_1 = {2, 4, 6}

7. Simple Event:- Event having one sample point.

E = {H}

8. Compound Event:-

Event having more than sample point

E_1 = {2, 3, 5}

9. Equally likely or Equiprobable :- Two outcomes are called equally likely if occurrence of one is not favourable over other.

P(H) = 1/2 P(T) = 1/2

if P(A) = P(B)

⇒ A & B are equilikely outcomes

10. Set of Exhaustive Event:- If all possible outcomes are covered by a set of events then this set is called set of exhaustive event.

Sample space = {1, 2, 3, 4, 5, 6}

Examples :-

Die is rolled → {1, 2, 3, 4, 5, 6}

E_1 → outcome is even → E_1 = {2, 4, 6}

E_2 → outcome is odd → E_2 = {1, 3, 5}

E_3 → outcome is prime → E_3 = {2, 3, 5}

E_4 → outcome is multiple of 3 → E_4 = {3, 6}

$E_1 \cup E_2$ = {2, 4, 6, 1, 3, 5} = Sample space

E_1 & E_2 are called set of Exhaustive

E_1 & E_3 are not set of Exhaustive

** If $E_1 \cup E_2 \cup E_3$ = Sample space

⇒ E_1 E_2 & E_3 are called set of exhaustive.

11. Disjoint Event or Incompatible Event or Mutually Exclusive Events :-

Two events are called mutually exclusive if occurrence of one rules out chance of occurrence of other.

Example :-

Die is rolled → {1, 2, 3, 4, 5, 6}

E_1 → outcome is even → $E_1 = \{2, 4, 6\}$

E_2 → outcome is odd → $E_2 = \{1, 3, 5\}$

E_3 → outcome is prime → $E_3 = \{2, 3, 5\}$

E_4 → outcome is multiple of 3

→ $E_4 = \{3, 6\}$

$E_1 \cap E_2 = \phi \Rightarrow E_1$ & E_2 are set of mutually exclusive events

$E_1 \cap E_3 = 2 \Rightarrow E_1$ & E_3 are not set of mutually exclusive

$E_1 \cap E_3 \cap E_4 = \phi$

$\Rightarrow E_1, E_3$ & E_4 are set of mutually exclusive

✱✱

If $E_1 \cap E_2 \cap E_3 = \phi$.

then E_1, E_2 & E_3 are called set of mutually exclusive events

12. Independent Event :-

Two events are called Independent events if occurrence or non-occurrence of one does not affect the chance of occurrence of other.

Classical definition of Probability :-

Let a sample space contains 'm+n' equally likely outcomes out of which m-favours occurrence of an event 'E', then probability of occurrence of E is

$$P(E) = \frac{m}{m+n} = \frac{\text{No. of favourable outcomes}}{\text{Total no. of outcomes}}$$

Odds in favour of event = $\frac{\text{No. of favourable outcomes}}{\text{No of Unfavourable outcomes}}$

Odds against the event = $\frac{\text{No of Unfavourable outcomes}}{\text{No. of favourable outcomes}}$

3 → R
4 → G
6 → B

E_1 = Red Ball
odd in favour of $E_1 = 3/10$
odds in against of $E_1 = 10/3$

Conditional Probability :-

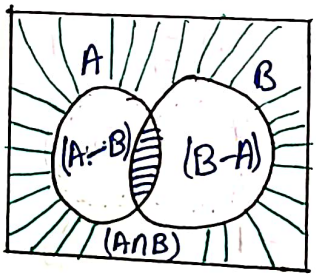
If E and F are two events associated with the same space of a random experiment are conditional probability of event E given that F has occurred.

$$P\left(\frac{E}{F}\right) = \frac{P(E \cap F)}{P(F)}, \quad P(F) \neq 0$$

$P(A/B) = P\left(\frac{A}{B}\right) = P(A/B)$ = Conditional probability of event A if event B has occurred already

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$ → Prob. of A if B occurred
given: B has occured

$$P(B/A) = \frac{P(B \cap A)}{P(A)}$$ → Prob. of B if A has occured only
given: A has occured.



Neither A Nor B
$A' \& B'$
$A' \cap B'$

$A \cup B = \underbrace{(A-B)}_{A} + \underbrace{(A \cap B)}_{B} + \underbrace{(B-A)}_{B} + \underbrace{(A \cap B)}_{A}$
$- (A \cap B)$
$- (A \cap B)$

$$A \cup B = A + B - (A \cap B)$$

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$\frac{n(A \cup B)}{n(U)} = \frac{n(A)}{n(U)} + \frac{n(B)}{n(U)} - \frac{n(A \cap B)}{n(U)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\frac{n(A' \cap B')}{n(U)} = \frac{n(U)}{n(U)} - \frac{n(A \cup B)}{n(U)}$$

$$P(A' \cap B') = 1 - P(A \cup B)$$

↓

Neither A nor B.

Demorgan's law :-

$(A \cup B)' = A' \cap B'$

$A' \cap B' = (A \cup B)' = U - (A \cup B)$

$A' \cap B' = U - (A \cup B)$

$n(A' \cap B') = n(U) - n(A \cup B)$

$$P(A' \cap B') = 1 - P(A \cup B)$$

2) $(A \cap B)' = A' \cup B'$

$A' \cup B' = (A \cap B)' = U - (A \cap B)$

$n(A' \cup B') = n(U) - n(A \cap B)$

$$P(A' \cup B') = 1 - P(A \cap B)$$

1. $n(A \cup B) = n(A) + n(B) - n(A \cap B)$
2. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
3. $P\left[\frac{A \cup B}{F}\right] = P\left(\frac{A}{F}\right) + P\left(\frac{B}{F}\right) - P\left[\frac{A \cap B}{F}\right]$

4. $n(A' \cap B') = n(U) - n(A \cup B)$

$P(A' \cap B') = 1 - P(A \cup B)$

↓

Neither A nor B

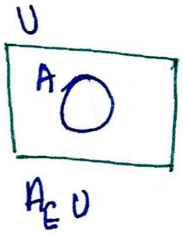
6. $n(A' \cup B') = n(U) - n(A \cap B)$

$P(A' \cup B') = 1 - P(A \cap B)$

5. $P(\bar{E}) = P(E') = 1 - P(E)$

$P(\bar{E}) = 1 - P(E)$

$P\left(\frac{\bar{E}}{F}\right) = 1 - P\left(\frac{E}{F}\right)$



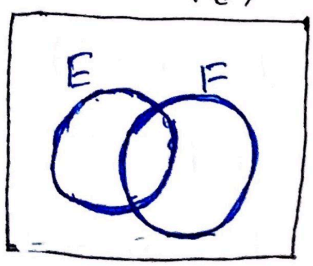
Properties of Conditional probability :-

Let E and F be events of a sample space S of an experiment, then

$$P(S/F) = P(F/F) = 1$$

1. $$P(S/F) = \frac{P(S \cap F)}{P(F)} = \frac{P(F)}{P(F)} = 1$$

2. $$P(F/F) = \frac{P(F \cap F)}{P(F)} = \frac{P(F)}{P(F)} = 1$$



Property :-

Let A and B are any two events of a sample space S and F is an event of S such that $P(F) \neq 0$, then $P[(A \cup B)/F] = P(A/F) + P(B/F) - P[(A \cap B)/F]$

$$P(E'/F) = 1 - P(E/F)$$

Multiplication Theorem in probability :-

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A \cap B) = P(B) \times P(A/B)$$

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

$$\boxed{P(A \cap B) = P(B) \times P(A/B)}$$

$$P(A \cap B) = P(A) \times P(B/A)$$

Prob of first event A

Prob. of second event B if A has occured.

$$P(A \cap B) = P(B) \times P(A/B)$$

$$P(A \cap B \cap C) = P(A) \times P(B/A) \times P(C/(A \cap B))$$

Note:- If A_1, A_2, A_3, A_4 are events, then

$$P(A_1 \cap A_2 \cap A_3 \cdots A_n) = P(A_1) \times P(A_2/A_1) \times P(A_3/A_1 \cap A_2) \times P(A_4/A_1 \cap A_2 \cap A_3)$$

Independent Events :-

Two events A and B are called independent if occurrence of one does not affect occurrence of other

$$P(A) = P(A/B)$$

$$P(A) = \frac{P(A \cap B)}{P(B)}$$

$$\Rightarrow P(A \cap B) = P(A) \times P(B)$$

**

If A & B are independent events, then

$$P(A \cap B) = P(A) \times P(B)$$

If event A is independent of event B

$$P(A/B) = P(A)$$

$$\frac{P(A \cap B)}{P(B)} = P(A)$$

$$P(A \cap B) = P(A) \times P(B)$$

$$\frac{P(A \cap B)}{P(A)} = P(B)$$

$$\Rightarrow P(B/A) = P(B)$$

$$P(A/B) = P(A)$$

$$\Rightarrow P(A/B) = P(A)$$

Note :-

1. If A and B are independent event then $P(A \cap B) = P(A) \times P(B)$

2. If A is independent of B then P will be independent of Q,
where P = A or $\bar{A}$ & Q = B or $\bar{B}$

RR

If A & B are independent events then

$\Rightarrow P(A \cap B) = P(A) \times P(B)$

a). $\bar{A}$ & B are also independent events

$\Rightarrow P(\bar{A} \cap B) = P(\bar{A}) \times P(B)$

b) $\bar{A}$ & B are also independent events

$\Rightarrow P(\bar{A} \cap B) = P(\bar{A}) \times P(B)$

c) A & $\bar{B}$ are also independent events

$\Rightarrow P(A \cap \bar{B}) = P(A) \times P(\bar{B})$

1 If A and B are independent events then

$$P(A \cup B) = P(A) + P(B) - P(A) \times P(B)$$

2. If A and B are mutually exclusive then

$$P(A \cup B) = P(A) + P(B)$$

1. A & B are independent

$$P(A \cap B) = P(A) \times P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) - P(A) \times P(B)$$

2. A and B are mutually Exclusive

$$A \cap B = \phi$$

$$P(A \cap B) = 0$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

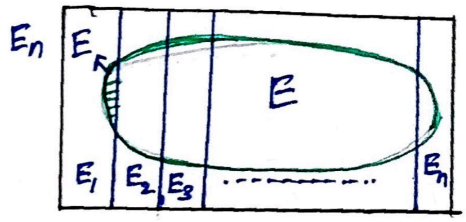
$$P(A \cup B) = P(A) + P(B)$$

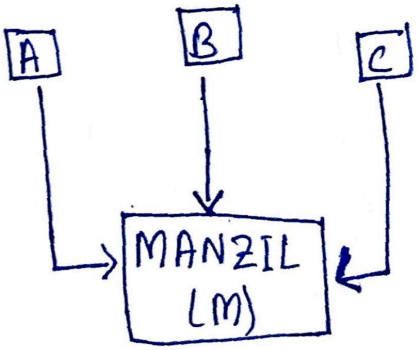
Total Probability Theorem:-

Let $E_1, E_2, E_3 \ldots E_n$ are n mutually Inclusive or disjoint and set of exhaustive events and E is also defined in the same space

$$E_1 \cup E_2 \cup E_3 \ldots \cup E_n = S$$

$$E_1 \cap E_2 \cap E_3 \ldots \cap E_n = \phi$$





$$P(\text{MANZIL}) = P(M) = P(A) \times P(M/A) + P(B) \times P(M/B) + P(C) \times P(M/C)$$

$P(A) \Rightarrow$ Prob. of selecting path-A

$P(M/A) \Rightarrow$ Prob of manzil if path A is selected

$P(B) \Rightarrow$ Prob of selecting path-B

$P(M/B)$ ⇒ Prob of getting M if B is selected

$P(C)$ ⇒ Prob of selecting path-C

$P(M/C)$ ⇒ Prob. of getting manzil if path-C is selected

Total Probability Theorem:-

$$P(E) = P(E_1) \times P\left(\frac{E}{E_1}\right) + P(E_2) \times P\left(\frac{E}{E_2}\right) + P(E_3) \times P\left(\frac{E}{E_3}\right) \cdots P\left(\frac{E}{E_n}\right)$$

$P(E_1)$ ⇒ Prob. of path-1 is used

$P\left(\frac{E}{E_1}\right)$ ⇒ Prob. of E if E_1 is used

Bayes Theorem :-

If $E_1, E_2 \cdots E_n$ are n non empty events which constitute a partition of sample space S, i.e $E_1, E_2 \cdots E_n$ are pairwise disjoint and $E_1 \cup E_2 \cup \cdots \cup E_n = S$ and A is any event of non-zero probability, then

$$P(E_i/A) = \frac{P(E_i)P(A_i/E_i)}{\sum_{j=1}^{n} P(E_j)P(A/E_j)} \quad \text{for any } i = 1, 2, 3 \cdots n$$

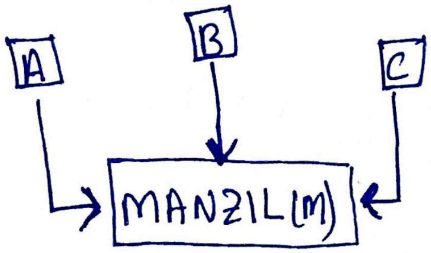
Proof :-

By formula of conditional probability, we know that

$$P(E_i/A) = \frac{P(A \cap E_i)}{P(A)}$$

$$= \frac{P(E_i)P(A/E_i)}{P(A)} \quad \text{(by multiplication rule of probability}$$

$$= \frac{P(E_i)P(A_i)}{\sum_{j=1}^{n} P(E_j)P(A/E_j)} \quad \text{by the result of theorem of total probability}$$



$$P(A/M) = \frac{P(A \cap M)}{P(M)} = \frac{P(A) \times P(M/A)}{P(A) \times P(M/A) + P(B) \times P(M/B) + P(C) \times P(M/C)}$$

$P(\text{MANZIL}) = P(m) = P(A) \times P(m/A) + P(B) \times P(m/B) + P(C) \times P(m/C)$

$P(m)$

contribution of path C

contribution of path – A

contribution of Path – B

$$P(A/m) = \frac{\text{contribution of A}}{P(m)} = \frac{P(A) \times P(m/A)}{P(m)}$$

$$P(B/m) = \frac{\text{contribution of B}}{P(m)} = \frac{P(B) \times P(m/B)}{P(m)}$$

$$P(C/m) = \frac{\text{contribution of C}}{P(m)} = \frac{P(C) \times P(m/C)}{P(m)}$$

Random Variable :-

A random variable is a real valued function whose domain is the sample space of a random experiment.

e.g:-

Tossing a coin two times in succession

Sample space = HH, HT, TH, TT

If x denotes the no. of heads obtained

$X(HH) = 2, X(HT) = 1, X(TH) = 1, X(TT) = 0$

If Y denotes the no. of heads minus the no. of tails

$Y(HH) = 2, Y(HT) = 1, Y(TH) = Y(TT) = 2$

2 coins are tossed

Sample space = {HH, HT, TH, TT}

X = Random Variable

X = No. of heads

$X(HH) = 2$

$X(HT) = 1$

$X(TH) = 1$

$X(TT) = 0$

Probability Distribution:-

It is a table or equation or ordered pairs which related every outcome with its probability of occurrence.

X = Random variable

x	$P(x)$
0	1/4
1	2/4
2	1/4

X = No. of heads

Sample space = {HH, HT, TH, TT}

$$\frac{1}{4} + \frac{2}{4} + \frac{1}{4} = 1$$

** Sum of probabilities of all the possible values of Random Variable is equal to 1

Let 3 coins are tossed

Sample space = {HHH, HHT, HTH, THH, TTT, TTH, THT, HTT}

X = No. of Tails = Radom variable

X	P(X)	
0	1/8	(HHH)
1	3/8	(HHT, HTH, THH)
2	3/8	(TTH, THT, HTT)
3	1/8	(TTT)

$1/8 + 3/8 + 3/8 + 1/8 = 1$

X	P(X)
x_1	p_1
x_2	p_2
x_3	p_3
x_4	p_4

$$p_1 + p_2 + p_3 + p_4 = 1$$

Mean or Expected Value:-

Let X be a random variable whose possible values $x_1, x_2, x_3 \ldots x_n$ occur with probability p_1, p_2, p_3, p_n resp. The mean of X denoted by μ is the number $\sum x_i p_i$ i.e the mean of X is the weighted average of the possible values of X, each value being weighted by its probability with which it occurs.

$$\text{Mean} = E(X) = \mu = \sum_{i=1}^{n} x_i p_i = x_1 p_1 + x_2 p_2 + \ldots x_n p_n$$

μ = mean or Expected value $= \sum_{i=1}^{5} x_i p_i$

$$= x_1 p_1 + x_2 p_2 + x_3 p_3 + x_4 p_4 + x_5 p_5$$

Variance of Random Variable :-

$$\sigma^2 = \sum_{i=1}^{n} p_i (x_i - \mu)^2$$

OR $$\sigma^2 = \sum_{i=1}^{n} p_i x_i^2 - \left[\sum_{i=1}^{n} p_i x_i\right]^2$$

σ^2 = Variance

$$\sigma^2 = \sum p_i x_i^2 - (\mu)^2$$

Standard Deviation = $\sigma = \sqrt{\text{variance}}$

Standard Deviation

$$\sigma_x = \sqrt{Var(x)} = \sqrt{\sum x_i p_i - (\sum x_i p_i)^2}$$